

Faculty vitae: Korosh Torabi

1. Education

Ph.D. Chemical Engineering Purdue University, West Lafayette, IN	Dec. 2011
M.S. Chemical Engineering Illinois Institute of Technology, Chicago, IL	July 2007
M.S. Energy Systems Engineering Sharif University of Technology, Tehran, Iran	June 2005
B.S. Chemical Engineering University of Isfahan, Isfahan, Iran	July 2003

2. Academic Experience

Clinical Assistant Professor University of Illinois Chicago, Chicago, IL Department of Chemical Engineering	Jan. 2023- Present
Assistant Professor Wayne State University, Detroit, MI Department of Chemical Engineering and Materials Science	Aug. 2014- Aug. 2022
Postdoctoral Fellow Northwestern University, Evanston, IL Department of Chemistry and Nonequilibrium Energy Research Center	Jan 2012-Aug 2014

3. Non-Academic Experience

Not applicable

4. Certification or professional registrations

None

5. Current membership in professional organizations

- o American Institute of Chemical Engineers
- o The Biophysical Society
- o American Chemical Society

6. Honors and Awards

- Not applicable

7. Selected service activities

Course development:

- Advanced Thermodynamics (UIC, CHE 501)
- Fluid Phase Equilibria (UIC, CHE 502)
- Transport Phenomena I (UIC, CHE 311)
- Transport Phenomena II (UIC, CHE 312)
- Chemical Reaction Engineering (UIC, CHE 321)
- Advanced Thermodynamics (WSU, CHE-MSE 7300)
- Thermodynamics (WSU, CHE 3300)
- Advanced Engineering Mathematics (WSU, CHE-MSE 7100)
- Materials Chemistry and Engineering (WSU, CHM 6060-7060)

Ph.D. Dissertations

Mojtaba Jokar, A statistical mechanical model for predicting stability and pairing-selectivity of coiled-coil protein structures, Aug. 2019. (PI, main advisor)

Zhixia Liu, Computational studies of allostery and dynamics of Myosin motor proteins, Expected April 2022. (PI, main advisor)

Master Theses

Deepak Joshi, A computational study of heterogeneous crystallization of nanoparticles via phase field crystal modeling, May 2017. (PI, main advisor)

Academic Administration and Service:

- o ChemE Connect committee, UIC
- o Department Action Team committee, UIC
- o College of Engineering Academic Operations Committee, WSU
- o College of Engineering Budget Committee, WSU
- o College of Engineering Academic Standards Committee, WSU
- o Undergraduate Curriculum and ABET Committee, WSU
- o Graduate Program Committee, WSU

8. Selected Publications and presentations

- o Kilani, M.; Torabi, K.; Mao, G. Application of Virtual Laboratories and Molecular Simulations in Teaching Nanoengineering to Undergraduate Students. *Computer Applications in Engineering Education* **2018**, 26 (5), 1527–1538.
- o Soroush Barhaghi, M.; Torabi, K.; Nejahi, Y.; Schwiebert, L.; Potoff, J. J. Molecular Exchange Monte Carlo: A Generalized Method for Identity Exchanges in Grand Canonical Monte Carlo Simulations. *The Journal of Chemical Physics* **2018**, 149 (7), 72318.
- o Zhou, Y.; Zhou, X.; Park, D. J.; Torabi, K.; Brown, K. A.; Jones, M. R.; Zhang, C.; Schatz, G. C.; Mirkin, C. A. Shape-Selective Deposition and Assembly of Anisotropic Nanoparticles. *Nano Letters* **2014**, 14 (4).
- o Torabi, K.; Schatz, G. C. Tensile Mechanics of α -Helical Polypeptides. *Macromolecules* **2013**, 46 (19), 7947–7956.
- o Torabi, K.; Corti, D. S. Toward a Molecular Theory of Homogeneous Bubble Nucleation: I. Equilibrium Embryo Definition. *The Journal of Physical Chemistry B* **2013**, 117 (41), 12479–12490.

- o Torabi, K.; Corti, D. S. Toward a Molecular Theory of Homogeneous Bubble Nucleation: II. Calculation of the Number Density of Critical Nuclei and the Rate of Nucleation. *The Journal of Physical Chemistry B* **2013**, *117* (41), 12491–12504.

Selected Conference Presentations, Seminars, and Invited Talks

- o Department of Chemical Engineering Seminar, April 2019. University of Toledo, Toledo, OH. Computational Studies of protein structure and dynamics.
- o Biophysical Society Annual Meeting, Winter 2018, San Francisco, CA. Zhixia Liu and Korosh Torabi, Allosteric Network in Myosin Motor Protein.
- o AIChE Annual Meeting, Fall 2017, Minneapolis, MN. Mojtaba Jokar and Korosh Torabi, Electrostatic Interactions and Coiled-coil Coupling Specificity.
- o 19th International Union of Pure and Applied Biophysics, Summer 2017, Edinburgh, UK. Korosh Torabi, Electrostatic Interactions and Coiled-coil Coupling Specificity
- o Biophysical Society Annual Meeting, Winter 2017, New Orleans, LA. Mojtaba Jokar and Korosh Torabi, Tensile mechanics of Coiled-coil peptides.
- o Biophysical Society Annual Meeting, Winter 2016, Los Angeles, CA. Mojtaba Jokar and Korosh Torabi, Tensile mechanics of Coiled-coil peptides.

9. Professional development activities

- The Biophysical Society Education Committee
- I have served as a member of the Education Committee of the Biophysical Society for a three-year term beginning July 1, 2019, and ending June 30, 2022.
- Engineering Technology Advisory Board at Henry Ford Community College
- I served as a member of the advisory board of the Engineering Technology at Henry Ford Community College from 2015 to 2018.
- AIChE Annual Meeting Session chair
- Nucleation and Growth I and III: AIChE Annual Meeting, Fall 2012, Pittsburgh, PA